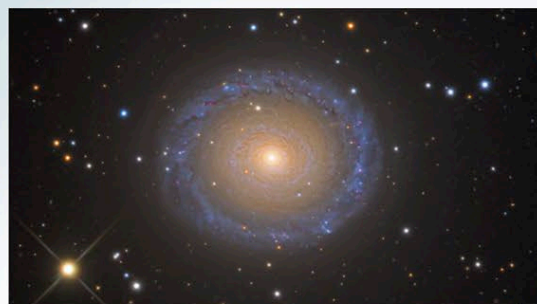
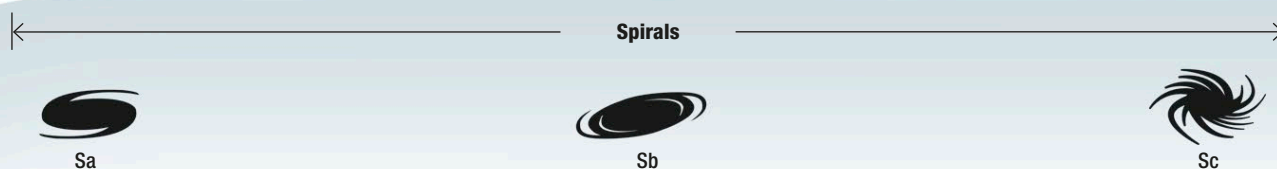
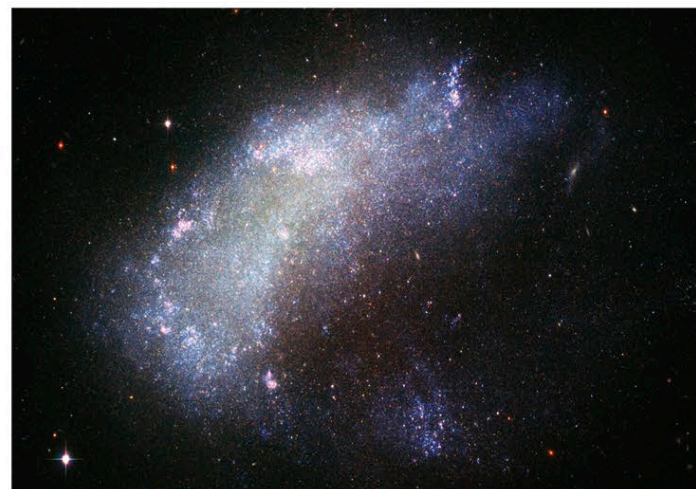


Irregular galaxies

Irregular galaxies are relatively shapeless clouds of gas, dust, and stars. The best known examples are the Large and Small Magellanic Clouds, which are the Milky Way's brightest satellite galaxies. Irregulars are rich in the raw materials of star formation and are often undergoing intense bursts of starbirth that make them bright for their size. Larger irregular galaxies show signs of some internal structure, such as central bars or lone, poorly defined arms. Hubble classified these as Irr I galaxies, compared to the truly shapeless Irr II irregulars.

▷ NGC 1427A

Dwarf irregular galaxies are thought to play an important role in galaxy evolution. Their stars have relatively few heavy elements, and probably represent raw material left over from the early history of the Universe, which has only recently ignited into star formation. NGC 1427A, for example, is lit up by newborn bright stars whose birth was triggered by its plunge into the Fornax galaxy cluster.



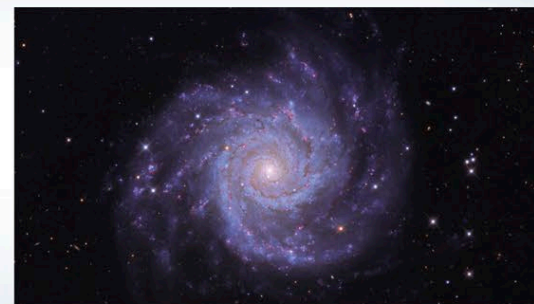
△ NGC 7217

Spiral galaxies of type Sa, such as NGC 7217 in Pegasus, have a central hub of older stars surrounded by a disk of stars and gas. Concentrated waves of star formation create tightly wound arms.



△ M91

Type Sb spirals have less tightly wound spiral arms emerging directly from the hub. M91 in Coma Berenices has relatively faint spiral arms for a galaxy of this type.



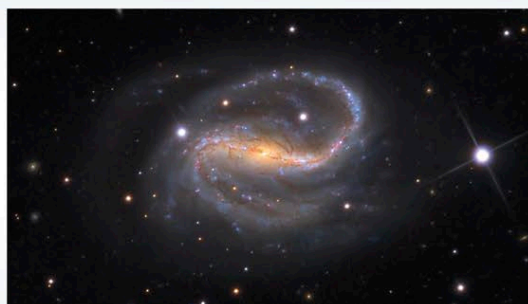
△ M74

Sc spirals such as M74 in Pisces have more loosely wound arms but are generally as bright as types Sa and Sb. The loosest Type Sd spirals, however, are usually a lot fainter.



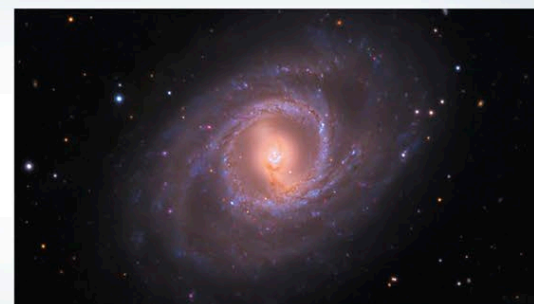
△ NGC 4921

Barred spirals follow the same general classifications as barless ones. SBa galaxies, such as NGC 4921 in Coma Berenices, have tightly wound spirals.



△ NGC 7479

Type SBb spirals, such as NGC 7479 in Pegasus, have a looser spiral structure but retain an obvious central bar emerging from either side of the nucleus.



△ M95

Type SBc galaxies have the loosest spiral arms, as seen in the beautiful M95, a barred spiral in some 38 million light-years away in Leo.